

Grade 3 Math Helper Hubs

★ Your Brain is a Muscle!

Math isn't about being fast or being perfect. It is about **growing your brain!**

Mistakes make you smarter!

When you do an easy math problem, your brain stays the same. But when you make a mistake and try to fix it, your brain actually builds new connections. Mistakes are proof that you are trying!

The Power of "YET"

If you forget how to do a problem during a review week, don't say, *"I don't know how to do this."*

Instead, add one magic word to the end: **"I don't know how to do this YET."**

Every time you try, your brain gets stronger!

★ Test-Day Superpowers

Big tests are just a chance to show off how much you've learned. Use these 3 superpowers to do your best!

Superpower 1: The Brain Breath

If a question looks scary, stop. Imagine you are holding a hot bowl of soup. Take a deep breath in through your nose to smell it, and blow out slowly through your mouth to cool it down. This gives your brain fresh oxygen so it can think clearly!

Superpower 2: The "3 Reads" Method

Don't just rush to the numbers! Read word problems three times:

- 1. 1st Read:** What is the story about?
- 2. 2nd Read:** What numbers do we need?
- 3. 3rd Read:** What math operation do we need to use?

Superpower 3: The Bullseye Approach

Aim for the very best answer on multiple choice!

- 1. Eliminate** obviously wrong options.
- 2. Check:** Does this option actually answer the question being asked?
- 3. Read carefully** to select the very best one.

★ Use Your Anchor Facts

Stuck on a hard multiplication fact? Don't just guess! Use an **Anchor Fact** (like your 2s, 5s, or 10s) to get close, then add the rest.

Example: What is 6×7 ?

Step 1: Use a 5s Fact

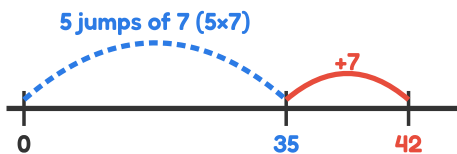
You might not know 6×7 , but 5s are easy!

$$5 \times 7 = 35$$

Step 2: Add One More Jump

You need 6 groups, so just add one more 7 to your anchor fact!

$$35 + 7 = 42$$

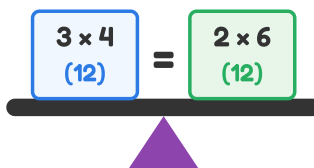


★ The Balance Scale

The $=$ sign does NOT mean "*the answer comes next.*" It means "**is the exact same amount as.**"

Both sides must balance!

If you see an equation like $3 \times 4 = 2 \times 6$, you have to do the math on BOTH sides to see if they weigh the same.



The left side is **12**.

The right side is **12**.

$12 = 12$, so the equation is

TRUE!

★ Pictures of Math

Multiplication builds a total. Division splits a total. Here is how we draw them!

Multiplication: Groups

3 groups of 4 = 12



Multiplication: Arrays

3 rows of 4 = 12



Division: Splitting the Total

12 total dots split into 3 groups = 4 in each group

$$12 \div 3 = 4$$



★ The Total Trick

Not sure whether to multiply or divide in a word problem? Look for the **TOTAL!**

Need to FIND the total? → Multiply.

If you have smaller pieces and want to know how many you have altogether, you multiply.

"Jon has 4 bags. There are 5 apples in each bag. How many apples does he have in all?"

4 × 5 = 20 Total Apples

Already HAVE the total? → Divide.

If the story gives you the big total number and asks you to share or split it, you divide.

"Jon has 24 apples. He puts an equal number into 3 bags. How many are in each bag?"

24 Total Apples ÷ 3 bags = 8 in each

★ Fact Family Triangles

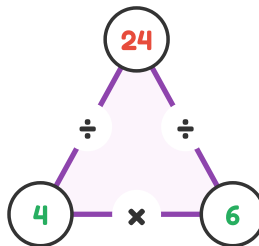
Multiplication and Division are connected! If you know one fact, you know the whole family.

The Fact Family Triangle

The big total lives at the top. The two smaller factors live at the bottom.

You can make 2 multiplication and 2 division equations from these three numbers!

$$\begin{array}{l|l} 4 \times 6 = 24 & 6 \times 4 = 24 \\ 24 \div 4 = 6 & 24 \div 6 = 4 \end{array}$$



★ Grouping & Breaking Apart

Use these properties to make tough multiplication problems easier!

Associative Property (Grouping)

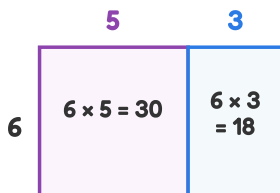
When multiplying 3 numbers, you can change the grouping but the answer stays the same!

$$(3 \times 2) \times 4 = 3 \times (2 \times 4)$$
$$6 \times 4 = 24 \quad 8 \times 3 = 24$$

Distributive Property (Breaking Apart)

To multiply 6×8 , you can break the tough 8 into two easier chunks: a 5 and a 3.

$$\begin{aligned} 6 \times 8 \\ &= 6 \times (5 + 3) \\ &= (6 \times 5) + (6 \times 3) \\ &= 30 + 18 \\ &= 48 \end{aligned}$$



★ The Rounding Hill

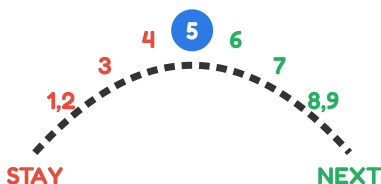
When rounding, find your target place value, then look at the digit to the right. Picture the Rounding Hill!

0, 1, 2, 3, or 4?

The number doesn't have enough energy. It rolls back down, so the target number **STAYS THE SAME**.

5, 6, 7, 8, or 9?

The number makes it over the top! It rolls forward, so the target number goes up to the **NEXT** one.



Example: Nearest 10

Round **348** to the nearest ten.

Target is 4. Look right at the 8.

8 is on the "NEXT" side.

The 4 goes up to 5 → **350**

Example: Nearest 100

Round **641** to the nearest hundred.

Target is 6. Look right at the 4.

4 is on the "STAY" side.

The 6 stays a 6 → **600**

★ Equation Balancing Tricks

You don't always have to do the big math! Try using these mental math tricks instead.

Trick 1: Decompose & Match

$$345 + 6 = 300 + [?]$$

Break apart 345. It is really just **300 + 45**.

So the left side is: **300** + 45 + 6 = **300** + [?]

Cross out the 300 on both sides. Now you just need to add 45 + 6 = **51**!

Trick 2: Too Much Subtraction!

$$435 - 98 = 435 - 100 + [?]$$

The problem tells us to subtract 100 because it's fast.

But wait, we only wanted to subtract 98! We subtracted 2 *too many*.

We have to put those 2 back. So the missing number is **2**!

★ Reading Number Lines

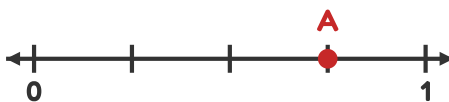
Fractions on a number line can be tricky! Follow these two steps to find any fraction.

Step 1: Find the Bottom Number (Denominator)

Count the total number of "jumps" it takes to get from 0 to 1. That is your bottom number!

Step 2: Find the Top Number (Numerator)

Start at 0 and count how many jumps it takes to get to your point. That is your top number!



4 total jumps to 1. Point A is at the 3rd jump. So A is at $\frac{3}{4}$!

★ Fraction Comparison Tricks

No visual model? No problem! Use these two tricks to compare fractions like a pro.

Trick 1: Same Bottom Numbers (Denominators)

If the bottom numbers are the same, the pieces are the exact same size. Just look at the top! **More pieces = Bigger fraction.**

$$\frac{5}{8} > \frac{3}{8} \quad (5 \text{ slices is more than } 3 \text{ slices!})$$

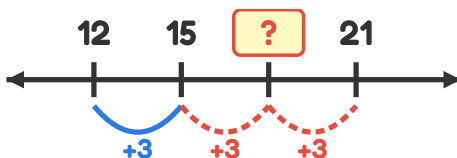
Trick 2: Same Top Numbers (Numerators)

If the top numbers are the same, you have the same number of pieces. But watch out: **A bigger bottom number means SMALLER pieces!** (Imagine sharing a pizza with 8 people instead of 3).

$$\frac{1}{3} > \frac{1}{8} \quad (1/3 \text{ slices are much bigger than } 1/8 \text{ slices!})$$

★ The Pattern Detective

When finding missing numbers in a pattern, look at the example and follow these 3 steps!



Step 1: Find Neighbors

Find two numbers next to each other. (12 and 15)

Step 2: Find the "Jump"

How much does it go up or down? ($12 + 3 = 15$)

Step 3: Test & Apply!

Use the jump to find the missing number! ($15 + 3 = 18$)

★ Word Problem Decoder

Don't just guess the math! Ask yourself: **What ACTION is happening in the story?**

+ Adding (Joining)

Are you putting different amounts together?

"Mia found 12 shells and 8 rocks. How many total?"

- Subtracting (Difference)

Are you taking some away, or finding how many are left?

"Sam had 50 pages, but read 20. How many are left?"

× Multiplying (Equal Groups)

Are you combining groups of the SAME size?

"There are 4 baskets with 6 eggs in EACH basket."

÷ Dividing (Sharing)

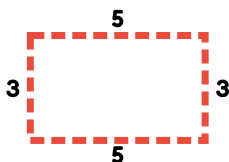
Are you splitting a big total into equal groups?

"30 markers shared EQUALLY with 5 friends."

★ Area vs. Perimeter

Perimeter = The Fence

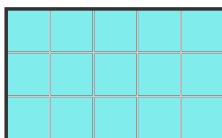
Perimeter is the total length of the **outside edge**.
Always add all the sides together!



$$5 + 3 + 5 + 3 = 16 \text{ units}$$

Area = The Grass

Area is the **inside space**. Multiply the length by the width to find the square units!



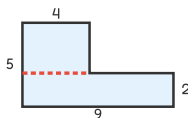
$$5 \times 3 = 15 \text{ square units}$$

★ The Chop & Add Strategy

When you need to find the area of a weird shape, follow these 3 steps!

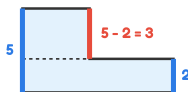
Step 1: Chop It!

Draw a line to chop the shape into 2 easy rectangles.



Step 2: Find the Hidden Side

Look at the parallel lines! The full left side is 5, but the bottom piece is 2. So the hidden top piece is $5 - 2 = 3$.

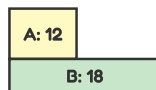


Step 3: Multiply & Add!

$$A = 4 \times 3 = 12$$

$$B = 9 \times 2 = 18$$

$$\text{Total} = 12 + 18 = 30 \text{ sq units}$$

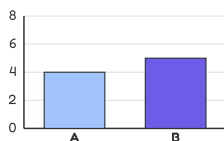


★ The Secret of the Scale & Key

Graphs are full of secrets! The biggest mistake is thinking that one line or one picture always means "1". You have to check the rules!

Bar Graph: Check the Scale!

Look at the numbers on the side. This graph is counting by **2s**! If a bar stops perfectly in the middle of 4 and 6, the secret number is **5**.



Picture Graph: Check the Key!

Never count pictures until you read the **KEY** at the bottom! If one circle equals 10 students, then 3 circles means **30**, not 3!

Grade	Number
3rd	

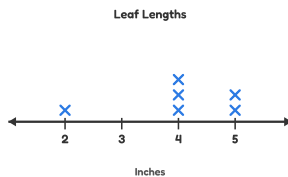
Key:  = 10 kids

★ X Marks the Spot (Line Plots)

Line plots can look confusing because there are numbers on the bottom and 'X's on the top. What do they mean?

The Line Tells You the SIZE

The number line at the bottom shows the measurement. In this example, it is the length of leaves in inches.



The 'X's Tell You HOW MANY

Every single 'X' represents one object. So, if there are three 'X's above the 4, it means **exactly 3 leaves** were 4 inches long!

**Count the total X's: $1 + 0 + 3 + 2$
= 6 total leaves measured!**

★ The Quadrilateral Family Tree

A **Quadrilateral** is any shape with 4 straight sides. But some shapes follow extra rules to get special names!



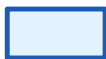
Quadrilateral

Any 4 sides



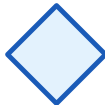
Parallelogram

Opposite sides are parallel



Rectangle

4 square corners
(L-shapes)



Rhombus

4 equal sides



Square

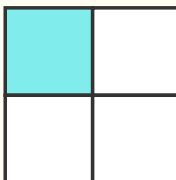
Follows ALL the
rules above!

Did you know? A square is very greedy. It has 4 equal sides, so it is a **Rhombus**. It has 4 square corners, so it is a **Rectangle**. And its sides are parallel, so it is a **Parallelogram**! It is all of them at once!

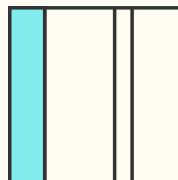
★ Equal Shares & Key Vocabulary

Equal Shares

To use a fraction, the shape MUST be cut into equal pieces!



Yes! ($\frac{1}{4}$)



No! (Unequal)

Parallel Lines

Lines that always stay the same distance apart and never touch!



Right Angle

A "square corner" that makes a perfect L-shape.



★ The Time Jumper

Time doesn't count to 100 like normal math... it only counts to **60**! Never try to "stack and add" time. Use a number line instead!

Reading the Clock



The **Short Hand** points to the hour. If it's between two numbers, the hour is the smaller number!

The **Long Hand** points to the minutes. Count by 5s for the big numbers, then by 1s!

Jumping on a Timeline

If you start at **1:45** and wait **35 minutes**...

1. Jump 15 minutes to reach the "safe" hour: **2:00**.
2. You have 20 minutes left to jump ($35 - 15$).
3. Jump 20 minutes to reach your answer: **2:20**!

★ Working Backwards & Multi-Step Time

Working Backwards

If you know the END time, jump backwards on your number line!

The game ends at 10:15. It lasted 45 minutes. When did it start?

1. Start at 10:15 and jump backwards 15 mins to **10:00**.
2. You have 30 mins left to jump ($45 - 15$).
3. Jump backwards 30 mins from 10:00 to reach **9:30**!

Multi-Step Time Tricks

Read word problems very carefully to see what they are actually asking for.

If a problem asks: "*How long did Carlos spend walking?*", make sure you don't count the time he spent sitting inside the store shopping! Only count the walking time.